



Tutorial & Programming S2B: Object Oriented Programming

Section A (Programming Questions)

1. A college library has books and CDs for loan. The items can be borrowed for three weeks. This table summarises the information to be stored.

Book	CD
Title of book	Title of CD
Author of book	Artist of CD
Book reference number	CD reference number
Whether it is on loan	Whether it is on loan
Date due for return of book	Date due for return of CD
Whether book is requested by another borrower	Type of music on CD (genre)

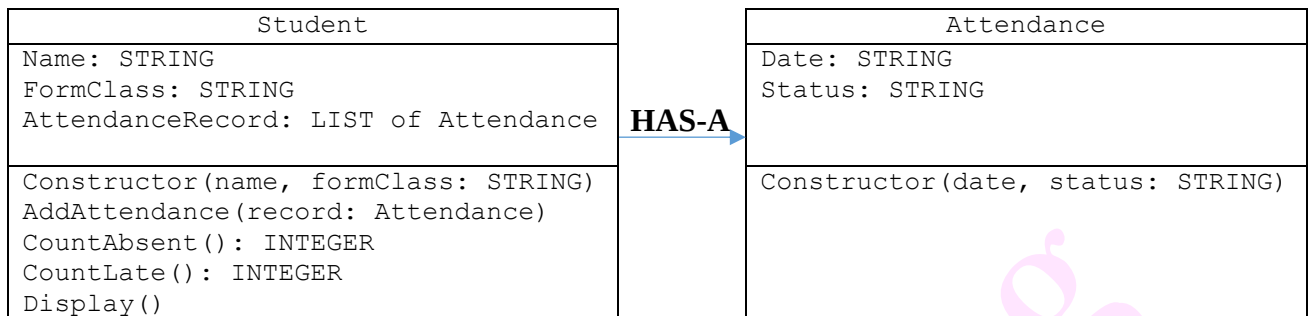
A base class **LibraryItem** and derived classes **Book** and **CD** are to be used to implement the library system. The class diagram for **LibraryItem** is defined as:

LibraryItem	
-Title	: STRING
-Author_Artist	: STRING
-ItemID	: INTEGER
-OnLoan	: BOOLEAN
-DueDate	: DATE
+Constructor() +GetTitle() +GetAuthorArtist() +GetItemID() +GetOnLoan() +GetDueDate() +Borrowing() +Returning() +PrintDetails()	

Implement the library system using OOP in Python, by writing code for the 3 classes.



2. Some classes share a **HAS-A** relationship as shown in the class diagram.



Status attribute may take one of the following: "PRESENT", "ABSENT", "LATE".

AttendanceRecord attribute is a list of Attendance objects.

AddAttendance method appends an Attendance object to the list.

CountAbsent method returns the number of times the student was absent.

CountLate method returns the number of times the student was late.

The following is a sample output from Display method:

"Johnny Lim, 23S99, was absent on 1 day(s) and late on 3 day(s)."

- (a) Write program code for the classes Student and Attendance.

Test your code by writing these lines of code:

```
a1 = Attendance("01/08/2021", "PRESENT")
a2 = Attendance("02/08/2021", "LATE")
a3 = Attendance("03/08/2021", "ABSENT")
student = Student("Johnny Lim", "23S99")
student.AddAttendance(a1)
student.AddAttendance(a2)
student.AddAttendance(a3)
absent = student.CountAbsent()
late = student.CountLate()
```

The text file `ATT.txt` stores a student's name and form class as the header and the attendance date and status in the remaining lines.

- (b) Write a function `AddRecords` to

- read data from the text file `ATT.txt`,
- instantiate a `Student` object from the header of the file,
- instantiate an `Attendance` object for each of the lines of data in the file,
- add each `Attendance` object to the `Student` object's `AttendanceRecord`, and
- return the `Student` object.

Test your function `AddRecords` to see if it works.

Sample data from `ATT.txt`:

```
ATT - Notepad
File Edit Format View Help
Johnny Lim, 23S99
01/08/2021, PRESENT
02/08/2021, PRESENT
03/08/2021, LATE
04/08/2021, LATE
05/08/2021, PRESENT
06/08/2021, LATE
100% Windows (CRLF) UTF-8
```



3. ACJC/Prelim/9569/2020/02/01

Write your program code in a single .ipynb file.

A programmer is writing a program to implement a role-playing computer game using Object-Oriented Programming (OOP).

The players have to collect food items. A food item has the following attributes:

- `name` : `STRING`
- `value` : `INTEGER`

and the following methods:

- `get_name()`
- `get_value()`

A player takes on the role of a person. A person has the following attributes:

- `name` : `STRING`
- `health` : `INTEGER` which is initialised at a value of 100
- `strength` : `INTEGER` which is initialised at a value of 100

and the following methods:

- `get_name()`
- `get_health()`
- `get_strength()`
- `eat(food)` adds the value of the food to the strength. The code should display the player's new strength.
- `attack(opponent)`

For the `attack` method, `opponent` is another person.

- A random integer `r` between 1 and 10 (inclusive) is generated.
- If the player's `strength` is less than `r`, then the player does not have enough strength to attack and there is no change to `opponent`'s health.
- If the player's `strength` is at least `r`, then the attack is successful and `opponent`'s health is decreased by `r`. If `opponent`'s health is now negative, then `opponent` has been defeated.
- The player's `strength` is decreased by `r`.

There are two subclasses of the `Person` class – `Healer` and the `Warrior`.

`Healer` has one additional method:

- `heal(patient)`

For the `heal` method, `patient` is another person.

- A random integer `r` between 1 and 10 (inclusive) is generated.
- If the player's `strength` is less than `r`, then the player does not have enough strength to heal and there is no change to `patient`'s health.
- If the player's `strength` is at least `r`, then the healing is successful and `patient`'s health is increased by `r`, up to a maximum of 100.



RAFFLES INSTITUTION H2 Computing (9569)

2025 Year 5

Warrior's attack method is twice as effective, meaning that if the player has enough strength to attack, opponent's health is decreased by $2 * r$, while the player's strength is decreased by r .

(a) Write program code to define the class `Food`. [3]

(b) Write program code to define the class `Person`.

The code should display appropriate messages about the outcome of `attack`, including the new value of opponent's health. [10]

(c) Use appropriate inheritance to write program code to define the class `Healer`.

The code should display appropriate messages about the outcome of `heal`, including the new value of patient's health. [4]

(d) Use appropriate inheritance and polymorphism to write program code to define the class `Warrior`. [2]

(e) Test your code with the following steps in order: [3]

- Create a `Food` item with name 'Cheese' and value 10.
- Create a `Warrior` with name 'Sam'.
- Create a `Healer` with name 'Alex'.
- Create a `Person` with name 'Jan'.
- 'Jan' attacks 'Sam'.
- 'Sam' attacks 'Jan'.
- 'Alex' heals 'Jan'.
- 'Sam' eats 'Cheese'.



Section B (Discussion Questions)

1. In the class definition for Book, add the missing instance variables and a procedure to *display* the book details.

```
CLASS Book
    #instance variables
    PRIVATE title
    #add missing instance variables here
    .....

    .....

    .....

    #constructor
    PUBLIC PROCEDURE new (aTitle, aAuthor, aISBN, aPrice)
        title = aTitle
        author = aAuthor
        isbn = aISBN
        price = aPrice
    ENDPROCEDURE

    PUBLIC PROCEDURE .....
        .....
    ENDPROCEDURE
ENDCLASS
```

2. A football club has two types of season ticket for their fans: pay-as-you-go and contract. All season ticket holders have their name, contact number and email address recorded. A pay-as-you-go ticket holder pays a chosen amount into their account. Each time the ticket holder buys a match ticket, the price of the ticket is deducted from the amount held in the account. The account can be topped up any time. A contract ticket holder pays a fixed fee per season. They can attend all the matches in the season.
 - (a) Draw a class diagram for the given three classes: SeasonTicketHolder, PayAsYouGoTicketHolder, ContractTicketHolder.
 - (b) Explain using an example what is meant by
 - (i) encapsulation;
 - (ii) instantiation of an object.
 - (c) In these three classes, attributes are declared as private while methods are declared as public. Explain why attributes are declared as private and methods as public.
 - (d) Write a statement to create a new instance of ContractTicketHolder with identifier as newCustomer, for name Johnny, contact number 91234569, email address johnnyyy@xyz.sg, contract fee \$999.



3. RVHS/Prelim/9597/2017/02/05

You are tasked by the school to write an application, using object-oriented programming language, which will store details of the library items inside the school library. There are currently two types of library items, namely books and media resources. The books have an ISBN number and the media resources have their digital capacity measured in GB stored. Common properties of a library item include:

- Library Item ID
- Title
- Author
- Description

(a) Draw a UML class diagram that shows how the properties could be distributed amongst the classes. Include in your diagram any **inheritance** between the classes. Also indicate some of the methods that would be required. Identify one method which demonstrates **polymorphism**.

(b) A music teacher approaches the librarian and requested to include a set of Music CDs into the library's collection. The teacher told the librarian that the song titles and durations should also be recorded for easy reference by his music students. Explain how this affects the classes, properties and methods given in (a).

(c) Code reuse, also called software reuse, is the use of existing software/code, or software knowledge, to build new software/code, following the reusability principles. Explain how your solution demonstrates software reuse.

(d) Explain how data hiding is achieved in a class by **encapsulation**. Support your answers with examples in your solution.

4. NYJC/Prelim/9597/2019/02/05

Goods in the warehouse are divided into 2 main categories – Kitchen appliances (e.g. kettle, toasters and ovens) and Entertainment products (e.g. LCD television, mp3 players and gaming consoles). Each item has an item name, description, unit price and quantity on hand. Kitchen appliances have an item weight, packing volume and colour. Entertainment products have a serial number, country of manufacture and recommended retail price.

(a) Draw a class diagram of the above showing inheritance, their private attributes and public methods.

(b) In relation to the diagram in part (a), explain the terms:

- (i) Encapsulation;
- (ii) Inheritance;
- (iii) Polymorphism.



5. 9597/2014/02/05 A software developer is given the task of producing software for a college. The software will help to manage information about what students do after finishing at the college.

The destination of each student after college is classified in one of three possible ways:

- University
- Employment
- Other

The college wishes to store:

- name
- number of A Level passes
- destination (U / E / O)
- university attended
- main subject studied at university
- type of employment
- what students do when their destination is classified as 'O'

The software developer will use an object-oriented approach to developing a solution.

(a) Draw a class diagram which exhibits the following:

- suitable classes with appropriate properties and methods
- inheritance
- polymorphism

[6]

(b) Explain how your solution to (a) demonstrates software reuse.

[2]

The data on the students is to be stored in a serial text file called `STUDENT.DAT`. Each line of the file has the same structure:

`<Name><NoOfPasses><Destination><University><MainSubject><EmpType><Other>`
with the string `NULL` stored where appropriate.

(c) Write an algorithm, in pseudocode, to read data from `STUDENT.DAT` and to output the following:

- total number of students going to university
- average number of passes for the students going to university
- total number of students

[7]



Section C (Assignment Question)

1. **9597/SP/02/05** A programmer is going to write an application, using an object-oriented programming language, which will store details of the vehicles available for hire from a vehicle rental company. The company hires out both cars and vans. The conditions of hire state that cars can only carry passengers and vans can only carry loads. Properties identified include:

- Registration code
- Maximum number of passengers
- Maximum load weight
- Type of vehicle

(a) Draw a diagram that shows how the properties could be distributed amongst a number of classes. Include in your diagram any inheritance between the classes. Also indicate some of the methods that would be required. One method should demonstrate polymorphism. [6]

(b) The company decide that they wish to also rent out minibuses for the carrying of passengers. They decide to offer 6, 12 or 14 seater minibuses. Explain how this affects the classes, properties and methods given in (a). [5]

(c) Explain, using this example, why encapsulation is an example of good programming practice. [3]



2. **9597/2016/02/06** A real-estate management company owns a number of residential and business units. When the company first acquires a unit, it often requires renovation work. The company records the renovation cost.

A residential unit will be a house or individual flat within a building. A business unit will be either an office building, a storage unit or a factory.

A residential unit is either advertised for sale, with the company looking to make a profit, or retained for rental. If the unit is sold, the sale price is recorded. If the unit is retained, the monthly rental charged, the start date and length of the rental (in months) are recorded.

A business unit has a long term lease, which is usually 10 years or longer. The company records the nature of the business. It does not offer any of its business units for sale.

Other data recorded for a unit include: purchase price, purchase date, number of rooms, floor space, whether or not a lift is present. The company records whether the house has a garage and whether it has a garden.

A programmer will develop an application, using object-oriented programming to store and process the company's data.

- (a) Draw a class diagram, with base class `UNIT`, showing:

- appropriate sub-class(es)
 - inheritance
 - the properties required
 - appropriate methods, including **one** pair of 'get' and 'set' methods for **one** of the properties.
- [8]

- (b) The company has recently purchased a number of units that they want to renovate as a 'block of flats' (a number of self-contained flats in the same building). Once the renovation is complete, the company may offer a block of flats for sale. Alternatively, it may retain the unit and advertise each individual flat for rental.

Explain how this would affect the design in **part (a)**. [3]

- (c) (i) Explain the meaning of the term encapsulation. [2]
(ii) Explain the meaning of the term polymorphism. [2]